# Week 01 Revision Questions

1. Do I *need* to set up Rust on my own computer, or can I just use VLAB / CSE?

   How could I go about setting up Rust on my own computer?

2. Let's write our first Rust program, hello world!

   First we'll write it as a `main.rs` file, and use the `rustc` command to compile it.

   Then, we will modify our program to be a `cargo` project instead.

   What are the differences in the two approaches? When should we use each approach in COMP6991?

3. Translate the following C program into Rust:

   ```
   #include <stdio.h>

   int main(void) {
       int my_integer = 42;

       if (my_integer > 0) {
           printf("%d is positive\n", my_integer);
       } else if (my_integer < 0) {
           printf("%d is negative\n", my_integer);
       } else {
           printf("%d is zero\n", my_integer);
       }

       return 0;
   }
   ```

4. Translate the following C program into Rust:

   ```
   #include <stdio.h>

   int main(void) {
       int count = 0;

       while (count < 10) {
           printf("%d\n", count);
           count++;
       }

       return 0;
   }
   ```

5. Translate the following C program into Rust:

   ```
   #include <stdio.h>

   int main(void) {
       for (int count = 0; count < 10; count++) {
           printf("%d\n", count);
       }
   }
   ```

   Is it possible to translate this approach directly to Rust? Why / why not?

6. Translate the following C program into Rust:

```c
#include <stdio.h>

void my_function(int *x);

int main(void) {
    int x = 42;

    my_function(NULL);
    my_function(&x);

    return 0;
}

void my_function(int *x) {
    if (x == NULL) {
        printf("no value for x\n");
    } else {
        printf("x has value %d\n", *x);
    }
}
```

Is this program possible to write in Rust? Discuss whether this is an advantage or disadvantage of Rust.